RESEARCH IMPLEMENTATION RECORD

Vehicle Tracking + Occlusions Research Project

Phase 1D and Phase 1E

Representative-event validation and recovery-aware evaluation framework

Purpose of this document

A precise continuation record of the work performed after Phase 1C. It documents how the planned Phase 1D representative-event validation was already satisfied by the Phase 1C execution sequence, and how Phase 1E converted the baseline outputs into recovery-aware measurements, candidate-retention analysis, manual outcome auditing, and consolidated comparison tables. The record is written to support later methodology, experimental-design, results, and limitations sections of the final research paper.

RECORD	STATE AS OF 19 OCTOBER 2026
Phase 1D	Satisfied by the representative-first execution already performed during Phase 1C; no duplicate re-run required
Phase 1E	Metric definitions, reference candidate generator, event-aware loss/recovery evaluator, manual audit, and final 24-run summary completed
Baseline trackers	SAM2.1 Tiny and DAM4SAM Tiny
Pilot events	E001-E012 from the frozen UA-DETRAC Phase 0 event inventory
Key caution	Phase 1E candidate retention uses annotation-backed candidates; it is not detector candidate recall or the proposed Phase 2 identity decision system

Source basis. Finalized Working Research Proposal v2 (September 2026), Phase 0 Frozen Documentation (28 September 2026), the Phase 1A-1C implementation record, and the actual Phase 1D/1E commands, CSV outputs, terminal summaries, and manual frame reviews produced in the project. No additional external experiment is claimed in this document.

Document map and evidence conventions

This is a continuation record, not a rewrite of Phase 0 or Phase 1A-1C. Only the inherited research question and the minimum baseline context needed to interpret Phase 1D and Phase 1E are repeated.

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Evidence labels used throughout

LABEL	MEANING
Measured	Direct output from generated CSV/JSON/terminal summaries.
Derived automatic	A result produced by the frozen Phase 1E rules from frame-level evaluator outputs.
Visually reviewed	Observation from tracker-overlay inspection performed during the project.
Limited / provisional	Useful evidence whose scenario boundary or generality is not strong enough for a broad research claim.

Reading rule

A tracker bounding box that overlaps an XML annotation is not automatically evidence of post-occlusion identity recovery. Phase 1E therefore separates continuous tracking, automatic loss/recovery signals, switch warnings, candidate retention, and manual interpretation.

1. Research goal and why Phase 1D/1E exist

Working research title: Reliability-Aware Sequential Identity Evidence Acquisition for Post-Occlusion Recovery of User-Selected Road Vehicles.

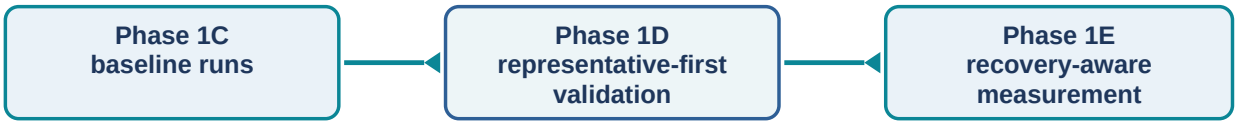
Frozen primary research question

Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

Phase 1D and 1E are baseline-enabling stages. Phase 1D checks the baseline pipeline first on representative failure conditions before broad execution. Phase 1E ensures that the baseline is measured in terms that are aligned with the research question: loss, identity ambiguity, recovery, false decisions, candidate availability, distractor burden, and computational cost - rather than only framewise overlap.

2. Phase 1D - Representative-first baseline validation

The Phase 1 plan specified that baseline testing should begin with four representative Phase 0 events rather than immediately sweeping every configuration over the full pilot set. In practice, this requirement was already satisfied during the Phase 1C execution sequence. Therefore, Phase 1D did not require a second duplicate experiment run.



Representative events were not re-run as a separate benchmark;
their intended role was already satisfied inside the Phase 1C execution sequence.

2.1 Representative events selected by the plan

EVENT	WHY IT WAS REPRESENTATIVE	VALIDITY NOTE
E011	Moving taxi with a visually similar taxi distractor; strongest initial same-type ambiguity case.	Important distractor case, but manual review showed tracking continuity rather than confirmed full disappearance.
E012	Same selected taxi under repeated occlusion and similar distractors; later ambiguity near scene exit.	Useful for false-continuation / switch-warning behavior; no verified reappearance frame in the frozen log.
E003	Provisional moving full-occlusion/reappearance candidate.	Full disappearance was not visually verified; later classified as continuous tracking under the event-aware rule.
E005	Long/repeated occlusion of a stationary SUV.	Useful for loss/resumption logic, but not evidence of hard moving-target recovery because the target is stationary.

2.2 How Phase 1D was actually satisfied

The baseline adapters, runner, evaluator, and visualization workflow were first exercised on these representative conditions, then expanded to all E001-E012. By the end of Phase 1C, SAM2.1 Tiny and DAM4SAM Tiny had each been run on all 12 events, yielding 24 tracker-event runs. This exactly fulfilled the intended Phase 1D progression: representative cases first, full pilot expansion only after the pipeline was functioning.

> **Why no additional Phase 1D re-run was performed**
>
> Re-running E011, E012, E003, and E005 under the same tracker versions, event ranges, and evaluator would only duplicate measurements already produced in Phase 1C. Phase 1D is therefore recorded as satisfied-by-execution, not as a separate new benchmark.

2.3 Connection to the research contribution

The representative-first strategy prevented the project from mistaking a stable tracker demo for identity-recovery capability. E011/E012 stressed visually similar distractors, E003 tested the boundary

between annotation-defined occlusion and visually verified disappearance, and E005 exposed the easier stationary-target case. These distinctions directly motivated the stricter Phase 1E measurement framework.

3. Phase 1E - Measure recovery rather than only tracking

Phase 1E formalized the measurements that the research actually needs. The central change was to stop treating a framewise target match as equivalent to post-occlusion recovery.

3.1 Frozen metric specification

MEASUREMENT	OPERATIONAL MEANING IN PHASE 1E
Pre-occlusion tracking	Whether the selected target was correctly tracked before the occlusion interval when pre-event evaluated frames are available.
Loss frame	First frame of a confirmed three-frame streak without a target match inside the relevant occlusion window.
ID switch	Not automatically declared. The evaluator only raises a switch warning; manual confirmation is required.
Correct recovery	After a detected loss, three consecutive target-matched frames beginning no earlier than the recorded reappearance frame.
False recovery	A distractor is accepted as the target. Phase 1E keeps this as a manual/identity-level judgment; warnings alone are insufficient.
No recovery	A verified reappearance exists but no valid post-loss recovery streak occurs.
Recovery delay	Recovery frame minus verified reappearance frame. Undefined for continuous tracking.
Candidate retention	Whether the true target survives the cheap plausibility gate in the annotation-backed reference candidate set.
Distractor count	Number of non-target plausible candidates remaining at the recovery frame.
Latency / FPS	Average per-frame tracker runtime and 1000 / latency_ms.

Core correction introduced by Phase 1E

No detected loss means no recovery metric. Continuous tracks are reported as continuous tracking, not as recovery with zero delay. This removed a misleading behavior of the earlier evaluator.

3.2 Automatic rule set



Loss search is restricted to the occlusion interval and, when available, must occur before recorded reappearance. Recovery is not counted for continuous tracks and cannot begin before the verified reappearance frame.

Ground-truth IoU is used only for offline evaluation; it is not an online deployment signal.

The target-match rule remained prediction present AND target IoU >= 0.30. The key refinement was event awareness: loss search is limited to the occlusion/recovery interval, and recovery cannot be counted before the recorded reappearance frame. This avoids using XML boxes during heavy occlusion to imply an impossible early recovery.

4. Annotation-backed reference candidate generation

Candidate recall was identified as the especially important Phase 1 quantity: if the true vehicle never enters the recovery candidate set, no later plate, Re-ID, local-feature, or sequential evidence-acquisition method can recover it. Phase 1E therefore implemented a reference candidate layer before any Phase 2 identity reasoning.



Evaluation question: did the candidate set retain the true target?

Important: target ID is used only after filtering to measure retention; it is not used by the gate.

This is annotation-backed reference candidate retention, not detector candidate recall.

4.1 Implementation

The script `src/recovery/annotation_candidate_generator.py` extracts all UA-DETRAC vehicle annotations present at a chosen frame, including target ID, vehicle type, bounding box, width, height, and area. Candidate identity is not used by the filter. The true target ID is checked only afterward to measure retention.

A high-recall plausibility gate compares each candidate with the user-selection reference box using area and aspect-ratio consistency. Vehicle type was inspected, but for E011 all 10 candidates were cars, so type alone did not reduce ambiguity.

4.2 First E011 test

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
E011 / 378	10	9	Yes - target ID 24	8

4.3 Full pilot reference-retention test

The first full-event pass used a maximum area ratio of 3.0 and retained the true target in 8/10 eligible events. E003 and E004 failed the gate. Diagnosis showed that their aspect ratios remained stable, but the target became much larger between user selection and reappearance:

EVENT	TRUE TARGET AREA RATIO	ASPECT-RATIO RATIO	WHY INITIAL GATE FAILED
E003	12.418	1.066	Valid target grew far beyond the initial max area ratio of 3.0.
E004	5.747	1.080	Same issue: large scale change, not shape inconsistency.

The maximum area ratio was relaxed to 15.0 while keeping the gate deliberately permissive. The goal at this stage is recall, not aggressive pruning. This threshold was chosen with margin above the largest observed valid ratio (12.418).

> **Validity caution**
>
> The gate was adjusted after observing failures on the same pilot events. Therefore 10/10 retention is a pilot-fitting result, not evidence that the threshold generalizes to unseen videos. Independent evaluation will still be required.

5. Candidate-retention results

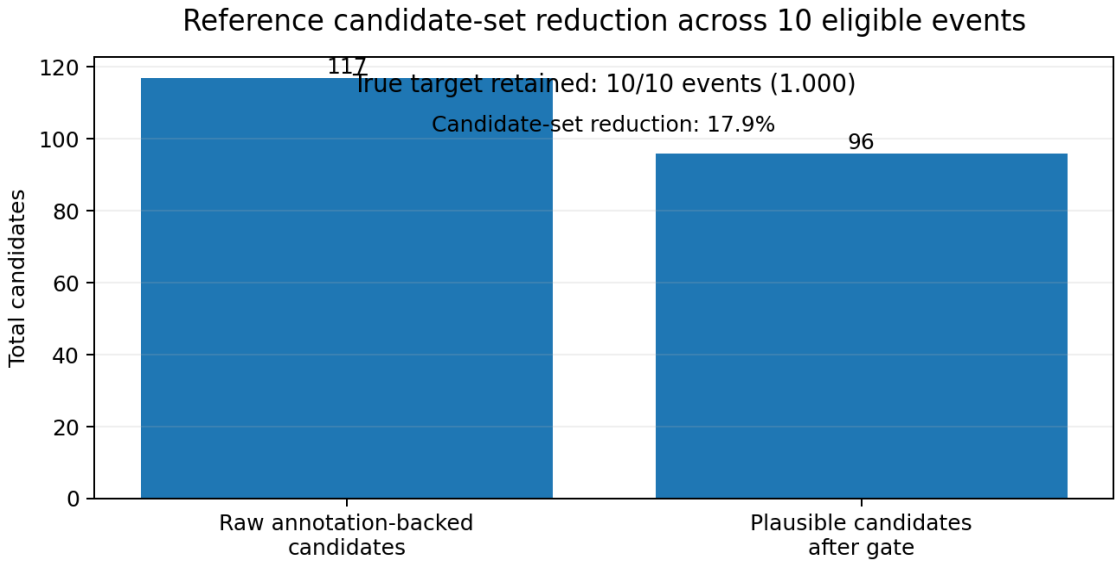


Figure 1. Annotation-backed candidate totals across the 10 events with a recorded reappearance frame.

EVENT	RAW	PLAUSIBLE	AFTER GATE ONLY	TARGET RETAINED
E001	12	12	11	Yes
E002	15	10	9	Yes

EVENT	RAW	PLAUSIBLE	DISTRACTORS	TARGET RETAINED
E003	11	10	9	Yes
E004	7	5	4	Yes
E005	11	8	7	Yes
E006	10	8	7	Yes
E007	9	7	6	Yes
E008	13	12	11	Yes
E009	19	15	14	Yes
E011	10	9	8	Yes

Measured reference result

Eligible events: 10. **True target retained:** 10/10 = 1.000. **Raw candidates:** 117. **Plausible candidates:** 96. **Reduction:** 17.9%. **Mean plausible candidates/event:** 9.60. **Mean distractors/event:** 8.60.

E010 and E012 were not included because the frozen event log has no recorded reappearance frame for them. For E010, later manual review also contradicted the original target-absence interpretation because the taxi remained partially visible at the end of the sequence.

6. Event-aware loss and recovery evaluator

Phase 1E created a new event-level evaluator over the existing `evaluation_frames.csv` outputs rather than rerunning either tracker. This preserved the Phase 1C predictions while correcting how recovery was interpreted.

6.1 Why the first automatic version was rejected

The initial rule searched for any three non-matching frames followed by any three matching frames. It produced impossible or misleading results: E005 recovered at frame 1044 even though the recorded reappearance was 1063; E006 recovered at 1143 although reappearance was 1249; E007 recovered at 1399 although reappearance was 1413. E003 was also labeled a no-recovery event because it lost overlap only after the recorded reappearance period.

> **Reason for the error**
>
> UA-DETRAC XML target boxes can continue through severe occlusion. Annotation availability is therefore not the same as visual availability. A recovery evaluator that ignores event boundaries can count matches inside an occlusion as recovery.

6.2 Corrected event-aware logic

The corrected evaluator restricts loss detection to frames at or after occlusion start and, when a reappearance frame exists, before that reappearance. Recovery search begins only at or after the recorded reappearance. If no loss occurs in the relevant interval, the run is classified as continuous tracking and recovery fields remain not applicable.

EVENT	SAM2.1	CSRT	INTERPRETATION
E003	Continuous; no loss	Continuous; no loss	Provisional full disappearance was not supported by the event-aware signal or manual review.
E005	Loss 869 -> recovery 1063	Loss 869 -> recovery 1063	Correct visible resumption of stationary SUV at recorded reappearance.
E006	Loss 1088 -> recovery 1249	Loss 1088 -> recovery 1249	Same stationary SUV; SAM2.1 also produced 24 switch warnings.
E007	Loss 1334 -> recovery 1413	Loss 1334 -> recovery 1413	Third overlapping stationary-SUV occlusion interval.
E012	Loss 598; no verified reappearance	Loss 602; no verified reappearance	Late ambiguous / boundary-exit case; recovery is not claimed.

6.3 Pre-occlusion tracking blanks

Some rows in the consolidated table have blank pre-occlusion tracking fields. These are not failures. For those runs, the stored frame-level evaluation begins at or after the event boundary, leaving no earlier evaluated frames from which to calculate a pre-occlusion tracking rate.

7. Manual outcome audit and identity-claim discipline

Automatic switch warnings were intentionally kept separate from confirmed identity switches. The manual audit records what was actually seen in tracker overlays without promoting heuristic warnings into stronger claims.

EVENT	MANUAL OUTCOME	CSRT-S	CSRT-R	CLAIM BOUNDARY
E001-E004	Continuous tracking	No confirmed switch	No confirmed switch	These are not post-loss recovery demonstrations.
E005	Loss and correct visible resumption	Correct SUV visible again	Correct SUV visible again; early predictions uncertain	Stationary target; useful recovery mechanics, limited generality.
E006	Loss and correct visible resumption	Incorrect predictions observed around 1107-1110; switch remains unconfirmed	Mostly abstained during ambiguous interval; resumed on correct SUV	Strongest qualitative difference between baselines in stationary occlusion.
E007	Loss and correct visible resumption	Correct SUV resumed; early narrow prediction uncertain	Same	Overlaps same stationary target/sequence as E005-E006.
E008-E011	Continuous tracking	No confirmed switch	No confirmed switch	E010 target-absence label invalidated by visual inspection.
E012	Ambiguous false continuation near scene exit	Late drift / 18 warnings	Fewer late warnings / 2 warnings	Neither confirmed ID switch nor false recovery is claimed without stronger visibility-aware ground truth.

Scientific reporting rule

Use phrases such as “switch warning”, “ambiguous false continuation”, or “incorrect prediction during occlusion” unless manual evidence establishes that the tracker actually accepted a specific distractor as the selected target.

This separation is important to the final research paper because the proposed contribution concerns false identity decisions. Overstating a geometric overlap warning as an identity switch would inflate the apparent baseline failure rate and weaken later comparisons.

8. Consolidated Phase 1E baseline results

METRIC	SAM2.1 TINY	DAM4SAM TINY
Tracker-event runs	12	12
Continuous-tracking runs	8	8
Runs with automatic loss	4	4
Automatic correct recoveries	3	3
Loss with no verified reappearance	1	1
Total switch-warning frames	42	2
Mean latency	1085.166 ms/frame	1109.282 ms/frame
Mean FPS	0.922	0.902

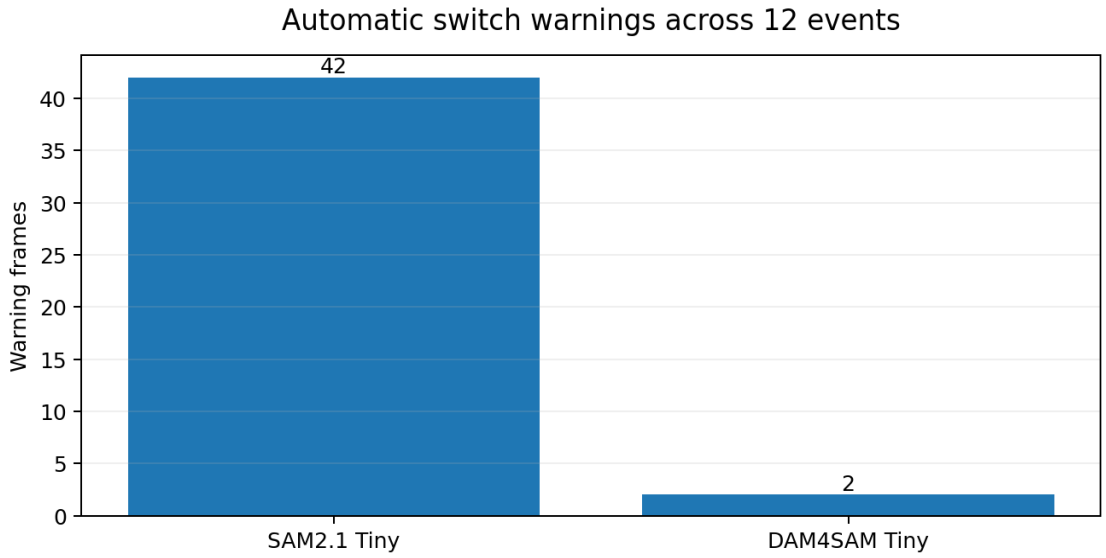


Figure 2. Automatic switch-warning frames accumulated across the 12 pilot events. Warnings are not confirmed identity switches.

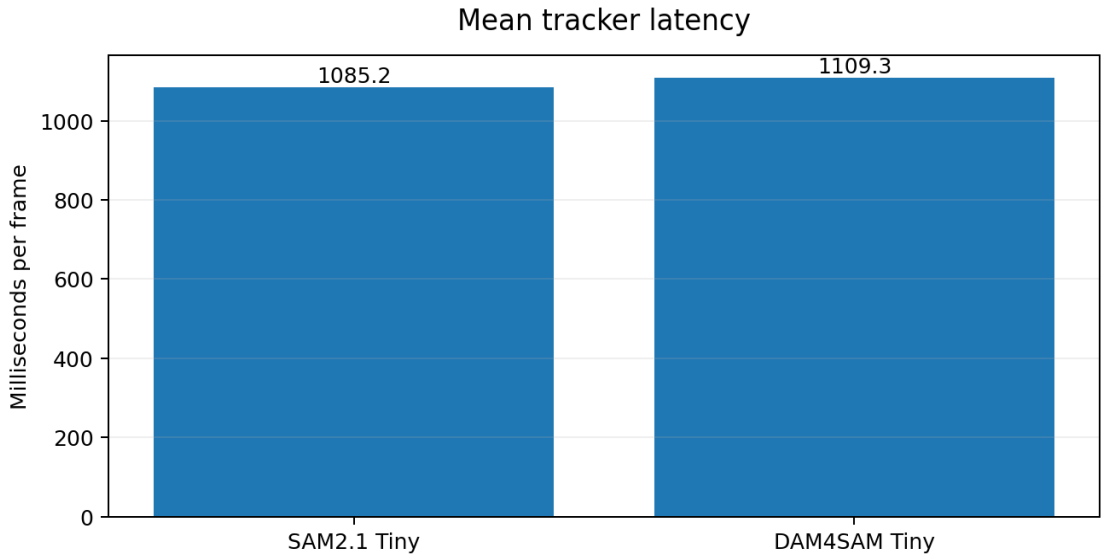


Figure 3. Mean measured per-frame tracker runtime on the project hardware/software setup.

9. What Phase 1D/1E established for the research

The main outcome is not that one tracker “wins” the research problem. Instead, Phase 1D/1E established a usable baseline measurement framework and clarified where the real identity-recovery problem begins.

9.1 Baseline behavior

Tracking continuity dominates the pilot set. Eight of the twelve events were classified as continuous tracking for both trackers under the corrected event-aware rule. These events remain useful for robustness checks but must not be counted as successful post-loss recovery.

Loss/resumption is concentrated in E005-E007. Both trackers showed automatic loss followed by correct visible resumption at the recorded reappearance frames. However, all three events concern the same stationary SUV in overlapping footage, so they are not three independent demonstrations of hard moving-vehicle identity recovery.

E012 exposes late ambiguity. SAM2.1 produced 18 switch warnings and DAM4SAM 2. The manual review found late ambiguity / false continuation near the scene boundary, but no verified reappearance exists, so recovery is not claimed.

DAM4SAM is not simply declared “better”. It produced far fewer switch warnings in these pilot runs, especially E006 and E012, while SAM2.1 had slightly lower mean latency. The pilot size, correlated events, and warning semantics prevent a broad superiority claim.

9.2 Candidate-generation implication

The annotation-backed reference candidate layer retained the true target in every eligible pilot event after the scale gate was relaxed. This demonstrates that the downstream identity stage can, in principle, be supplied with the true target under the pilot reference setup. The remaining candidate sets are still large (9.60 plausible vehicles/event on average), which is exactly where Phase 2 identity evidence and sequential cue acquisition become necessary.

> **Direct connection to the proposed contribution**
>
> Phase 1 now supplies the conditions that Phase 2 needs: a selected target history, a loss/recovery event boundary, and a candidate set that should contain the true vehicle. Phase 2 can then decide MATCH, REJECT, or ACQUIRE MORE EVIDENCE using reliability, identity value, and computational cost.

10. Validity constraints and what must not be claimed

CONSTRAINT	IMPLICATION FOR THE FINAL PAPER
Annotation-backed candidates	The 1.000 candidate-retention value is a reference retention result, not detector candidate recall. A real detector may miss vehicles.
Pilot-tuned area threshold	max_area_ratio=15.0 was chosen after diagnosing E003/E004; unseen-video generalization is untested.
XML visibility mismatch	Annotation presence does not prove the vehicle is visually available; event-aware rules and manual review remain necessary.
Correlated E005-E007	These are overlapping intervals of the same stationary target and must not be treated as three independent hard-recovery samples.
E003 provisional label	The planned “full occlusion” interpretation was not visually confirmed; Phase 1E classifies the run as continuous tracking.
E010 scenario correction	The target remained partially visible, so the Phase 0 “target absent” coverage is not supported by this event.
E012 no verified reappearance	Loss and ambiguity can be reported; correct/no recovery after true reappearance cannot be established from this event.
Switch warnings	42 vs 2 is useful diagnostic evidence, but warning frames are not confirmed identity switches.
Runtime environment	~0.9 FPS reflects the current research environment and adapters; it is a baseline implementation cost, not an optimized deployment speed.

10.1 Remaining evidence gaps inherited from the pilot

The scenario audit from Phase 1C remains relevant. A stronger final evaluation still needs carefully verified cases for: (i) a moving vehicle that fully disappears and later reappears among similar distractors, (ii) a true target-absence window with plausible distractors, and (iii) distractors that appear before the target returns. These gaps should be filled before making strong claims about the final recovery system.

These limitations do not invalidate Phase 1D/1E. They define the boundary of what the pilot evidence can support and guide the construction of later evaluation cases.

11. Implementation artifacts produced in Phase 1E

The following files form the reproducible Phase 1E record and should be retained with the repository:

ARTIFACT	ROLE
configs/phase1/phase1e_metrics.yaml	Frozen metric definitions and automatic event-aware rules.
src/recovery/annotation_candidate_generator.py	Annotation-backed reference candidate extraction and high-recall size/shape gate.
src/evaluation/evaluate_candidate_recall.py	Evaluates reference target retention and distractor burden at recorded reappearance frames.
results/phase1/phase1e_candidate_recall.csv	Per-event raw/plausible candidate counts and target-retention results.
src/evaluation/phase1e_event_metrics.py	Computes event-aware continuous tracking, loss, recovery, warnings, candidate metrics, latency, and FPS.
results/phase1/phase1e_event_metrics.csv	Automatic 24-run event-level metrics.
results/phase1/phase1e_manual_outcome_audit.csv	Manual interpretation layer kept separate from automatic warnings.
src/evaluation/build_phase1e_summary.py	Merges automatic and manual evidence into the authoritative Phase 1E result table.
results/phase1/phase1e_final_metrics.csv	Final consolidated 24-run Phase 1E table.
src/evaluation/summarize_phase1e.py	Produces tracker-level and candidate-level summary statistics.
results/phase1/phase1e_summary.txt	Final compact Phase 1E numerical summary.

11.1 Data lineage

Phase 1E does not alter the original Phase 0 frozen event log. It reads the same event definitions through the project event loader, reuses Phase 1C tracker predictions and frame-level evaluation files, and writes new Phase 1E analysis artifacts under `results/phase1/`. This preserves the provenance of the baseline runs while allowing stricter interpretation.

12. Phase 1D/1E freeze and handoff

Phase 1D status

Complete by absorption into Phase 1C. The four representative events were used to validate the baseline workflow before expansion to all E001-E012. No duplicate Phase 1D rerun is necessary.

Phase 1E status

Complete and ready to freeze as the pilot measurement framework. Recovery-aware metrics, annotation-backed candidate retention, event-aware loss/recovery rules, manual outcome auditing, and consolidated baseline summaries have been implemented and checked.

The frozen Phase 1E outputs should be used later as the baseline/protocol source when writing the final paper. In particular, the manuscript should preserve the distinctions between tracking continuity and recovery, warning and confirmed switch, reference candidate retention and detector recall, and stationary-target recovery and difficult moving-target recovery.

Phase 1E provides the measurement foundation for later work; it does not itself implement the proposed sequential identity-cue acquisition policy. That policy remains a subsequent stage.

Recommended paper reuse

FINAL PAPER SECTION	PHASE 1D/1E MATERIAL TO REUSE
Methodology / experimental protocol	Representative-first validation strategy; event-aware loss/recovery definitions; candidate-set construction; manual audit separation.
Baselines	SAM2.1 Tiny and DAM4SAM Tiny setup and 24-run pilot summary.
Metrics	Loss, recovery, delay, warning semantics, reference candidate retention, distractor count, latency/FPS.
Results	8 continuous events, 4 loss events per tracker, 3 automatic recoveries, E012 no verified reappearance, 42 vs 2 warning frames, latency comparison, 10/10 reference retention.
Limitations	Annotation-backed candidates, pilot-tuned gate, correlated stationary events, missing hard moving-target recovery / true absence cases.

End of Phase 1D-1E implementation record. This document intentionally stops before later Phase 1 components or the proposed Phase 2 identity-evidence policy.

Appendix A - Compact event-level Phase 1E result matrix

The table below is the compact manuscript-oriented view of the final event-level interpretation. It should be read together with the validity constraints above.

EVENT	CAM 1	CAM 2	CANDIDATE RETENTION	KEY INTERPRETATION
E001	Continuous	Continuous	Yes	Partially occluded initialization; no post-loss recovery.
E002	Continuous	Continuous	Yes	Severe partial occlusion; no detected loss.
E003	Continuous	Continuous	Yes	Provisional full-occlusion label not supported.
E004	Continuous	Continuous	Yes	Partial occlusion / boundary motion.
E005	Loss 869 -> rec. 1063	Loss 869 -> rec. 1063	Yes	Stationary SUV; correct visible resumption.
E006	Loss 1088 -> rec. 1249; 24 warnings	Loss 1088 -> rec. 1249; 0 warnings	Yes	Stationary SUV; strongest warning-count baseline contrast.
E007	Loss 1334 -> rec. 1413	Loss 1334 -> rec. 1413	Yes	Stationary SUV; overlaps E005-E006 footage.
E008	Continuous	Continuous	Yes	Moving target with distractors; no loss.
E009	Continuous	Continuous	Yes	Moving partial occlusion; no loss.
E010	Continuous	Continuous	N/A	Original target-absence scenario contradicted.
E011	Continuous	Continuous	Yes	Similar-taxi distractor case; continuity rather than recovery.
E012	Loss 598; 18 warnings; no verified reappearance	Loss 602; 2 warnings; no verified reappearance	N/A	Late ambiguity / false continuation near exit; no recovery claim.

Interpretation summary

The pilot baseline shows that the current trackers often maintain continuity, while the clearest loss/resumption cases are stationary-target occlusions. The Phase 1E framework therefore gives the later identity-recovery system a defensible measurement protocol without overstating what the current pilot proves.